

Where Nobody Lives

Building a national map of the census blocks that hold no people

Democracy's Data

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01 Where nobody lives

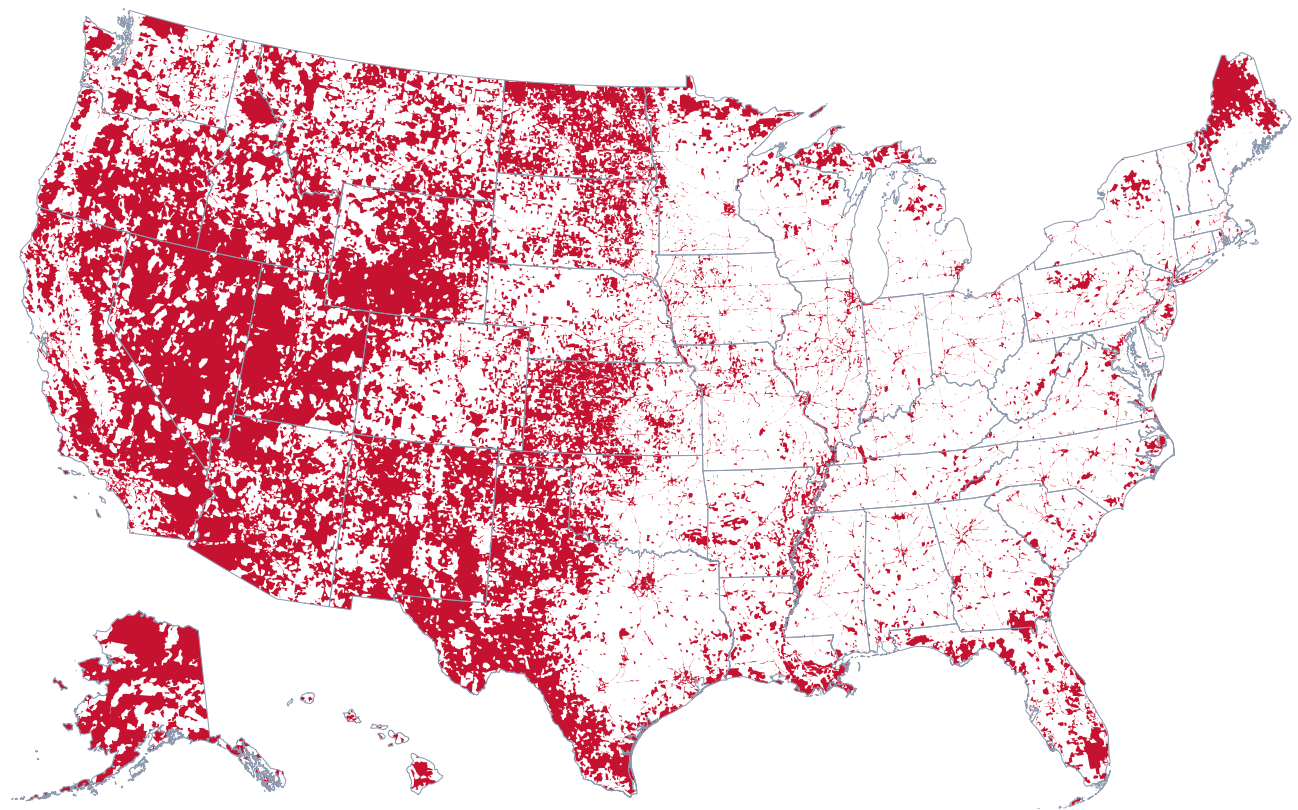


Figure 1. Every census block in the United States that recorded no residents in 2020.

Red is a block with nobody living on it, white is a block with residents. Click a state to zoom to it; click again, or press Escape, to come back. Hovering gives that state's population, its block count, the share of its blocks that are empty and the share of its land.

This is a map of where nobody lives in the United States. **41.5% of the country's land area** is in census blocks that recorded not one resident — an area of about 1,465,387 square miles — more than the combined land area of Alaska, Texas, California, Montana, New Mexico and Arizona, the 6 largest states in the country.

02 What a census block is

A census block is the smallest piece the Census Bureau cuts the country into, and the reason it exists is arithmetic. Every other geography the Bureau publishes — a tract, a city, a school district, a congressional district — has to be expressible as some pile of blocks, because counts add upward but cannot be divided downward. Make the smallest piece too big and whole classes of question stop having answers. The block is the Bureau's answer to how small the smallest piece has to be.

The Bureau draws them, not the states, and it draws them along things a person could stand next to: streets, streams, railroad tracks, shorelines, and the legal boundaries of counties, cities and reservations.¹ States do get a say. Through the Redistricting Data Program they can suggest where block boundaries should fall, so that blocks nest inside the precincts they actually run elections in. But the pen is federal, and a block is drawn for the convenience of counting rather than to mark anything a resident would recognize.

There are 8,174,955 of them and they are small. The median block with any land on it covers 455,841 square feet — about 10.5 acres — and holds 14 people.

The extremes run a long way from that median, in both directions and not together. The largest block in the country by area covers 9,226 square miles in North Slope Borough, Alaska, more land than 10 states and state equivalents including Vermont, whose 9,217 square miles it exceeds. Its population is 0: the biggest block in the country is an empty one. The largest by population is 8,727 people in Los Angeles County, California, on 38 acres.

That pairing is this brief's subject. 2,368,443 blocks — 29.0% of every block in the country — recorded nobody at all in 2020. That is not a defect in the count. It is what happens when you cut a country into pieces small enough that most pieces are a lake, an interchange, a runway, a cemetery, or a stretch of national forest.

This brief is a walkthrough. It shows where the data came from, what was done to it, and what the resulting map does and does not say. The point is not the map at the end; it is that you could rebuild it.

¹ The rule and the definition are in the Bureau's glossary entry for census block; the visible-feature principle is why a river or an interstate so often has blocks of its own on either side of it.

03 Where the data came from

One download, and it holds both halves of the problem.

TIGER/Line. The Bureau's TIGER/Line files are its digital outlines of every census unit. The layer used here is TABBLOCK20, the 2020 tabulation blocks, one zipped shapefile per state. There is no national file. To build the map you download each state separately, fifty-two files in all, about 10 GB for the country. Work through them one at a time: unzip a state, read the handful of columns you need, then delete what you unzipped before starting the next, so only one state is ever open. Keep the archives themselves and a second run costs nothing, since a file already on disk is not fetched twice. If space is short, they can go once the tables exist; nothing later reads them again.

The count rides along with the shape. Every block record in those files carries a POP20 field: the total population the 2020 decennial census counted in that block. It is not an estimate and not a sample. It is the decennial count, the same number published in the P.L. 94-171 Redistricting Data Summary File that every state legislature receives for re-

districting. The Bureau prepends it to the geography, so a block's boundary and its population arrive together, in one file, already joined.

The same counts are also served, one variable at a time, by the Census Data API, which has a chapter of its own. For a map of every block in the country it is much the slower road: blocks cannot be asked for a whole state at once, so covering the country takes some 3,100 requests against the 52 files downloaded here.

04 How the map was made

Four steps, and each one throws something away on purpose.

1. **Read** each state's block file and keep the blocks whose POP20 is zero. One state at a time, so that a single state is in memory rather than the whole country.
2. **Drop the water.** A block that is all water reports zero people truthfully, but drawing those fills the Great Lakes and every coastal bay with solid color. Blocks with no land area are set aside here, and counted separately.
3. **Dissolve** the surviving blocks into one shape. Touching empty blocks become one region, which is honest here: the boundary between two adjacent empty blocks is not a feature of the world, it is an artifact of where the Bureau put its lines. Every count in this brief is taken before this step, or carried through it as a summed field, because after a dissolve a count of shapes is a count of regions and not of blocks.
4. **Simplify** with mapshaper, project to Albers USA, and round the coordinates to whole units of the figure's own grid.

Simplification is the step that deserves suspicion, so here are its terms. The national layer at full resolution is about 200 MB, far too large to put in a web page. What ships is 2% of the original vertices: 999,532 points in 62,916 polygons, about 1 MB. At the width this figure is drawn, one unit of the grid is roughly 2.3 kilometers, so detail finer than that could not be seen even if it were kept. What is lost is the exact shape of any individual block. What survives is where the empty land is, which is the only claim the figure makes.

Two things in steps 3 and 4 are easy to get wrong and were got wrong here first. Dissolving produces polygons with **holes** — a populated town inside an empty county is an interior ring — and a ring drawn on its own is painted as fill rather than subtracted, which turns Nevada into a solid rectangle. Both renderings below therefore draw one path per polygon with all of its rings together, under an even-odd fill rule. And rounding coordinates has to happen *after* projecting, not before: rounding longitude and latitude to whole numbers rounds them to whole degrees, and the country dissolves.

The built tables are small and readable: `state_blocks.csv`, one row per state; `block_pop_hist.csv`, a count of blocks at each population from 0 to 1,000; `zero_land.csv`, land area of the empty blocks by state; `block_size.csv`, how big a block is and the two extremes; `dc_blocks.csv` with `dc_frame.csv` and `dc_facts.csv`, one city at full block resolution; `facts.csv`, the national totals quoted in this text; and the figure geometry in `map_zero.csv`, `map_states.csv` and `map_frame.csv`, stored as integer figure coordinates so the web version and the print version draw the same shapes from the same numbers.

Level	Where	Blocks	Blocks with nobody	Percent
Nation	50 states, DC, Puerto Rico	8,174,955	2,368,443	29.0%
Most empty	Alaska	28,568	16,803	58.8%
Least empty	Puerto Rico	41,987	5,417	12.9%

05 How big is a census block, anyway?

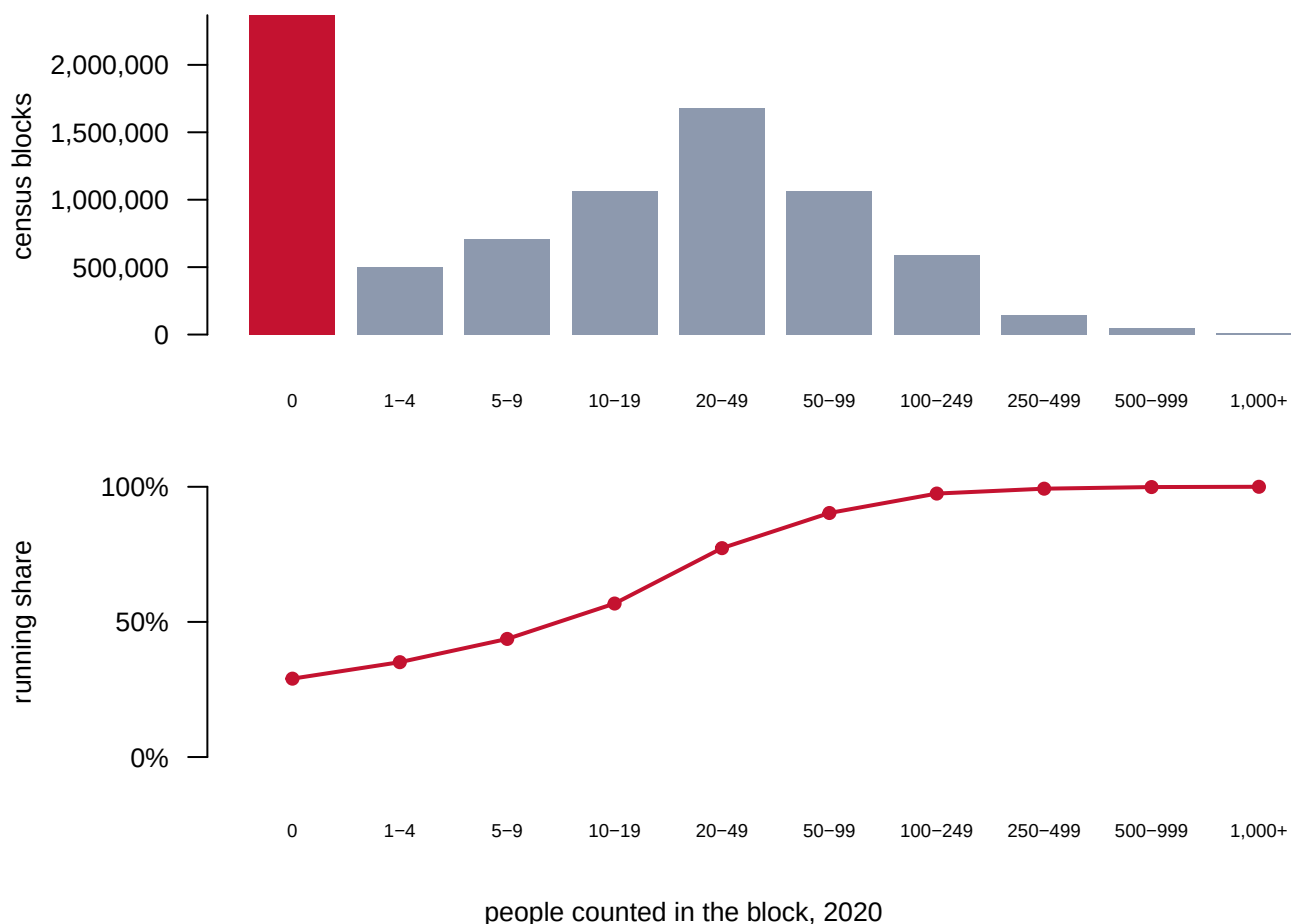


Figure 2. How many people live in a census block. Blocks grouped by population, from empty to a thousand and over; the red bar is the 2,368,443 with nobody. Bins widen toward the tail because block populations do: binned this way the whole range fits one axis that starts at zero, so every bar's length is its count and nothing has to be logged or broken. The panel beneath is the running share — the portion of all blocks at or below each bin. Hover a bin in the web edition for its exact count.

The shape says the block is a small thing. The median block in the country holds 14 people. Among blocks with anybody at all, the median is 28. 45.3% of all blocks hold ten people or fewer, and the running share passes half by the 20-49 bin. A census block is not a neighborhood; it is closer to a city block, which is what it was named for.

This matters for reading any block-level statistic. A percentage computed on a unit holding fourteen people is mostly noise, and the Bureau's disclosure-avoidance noise is delib-

erately largest, in relative terms, exactly here.

06 Which states are emptiest

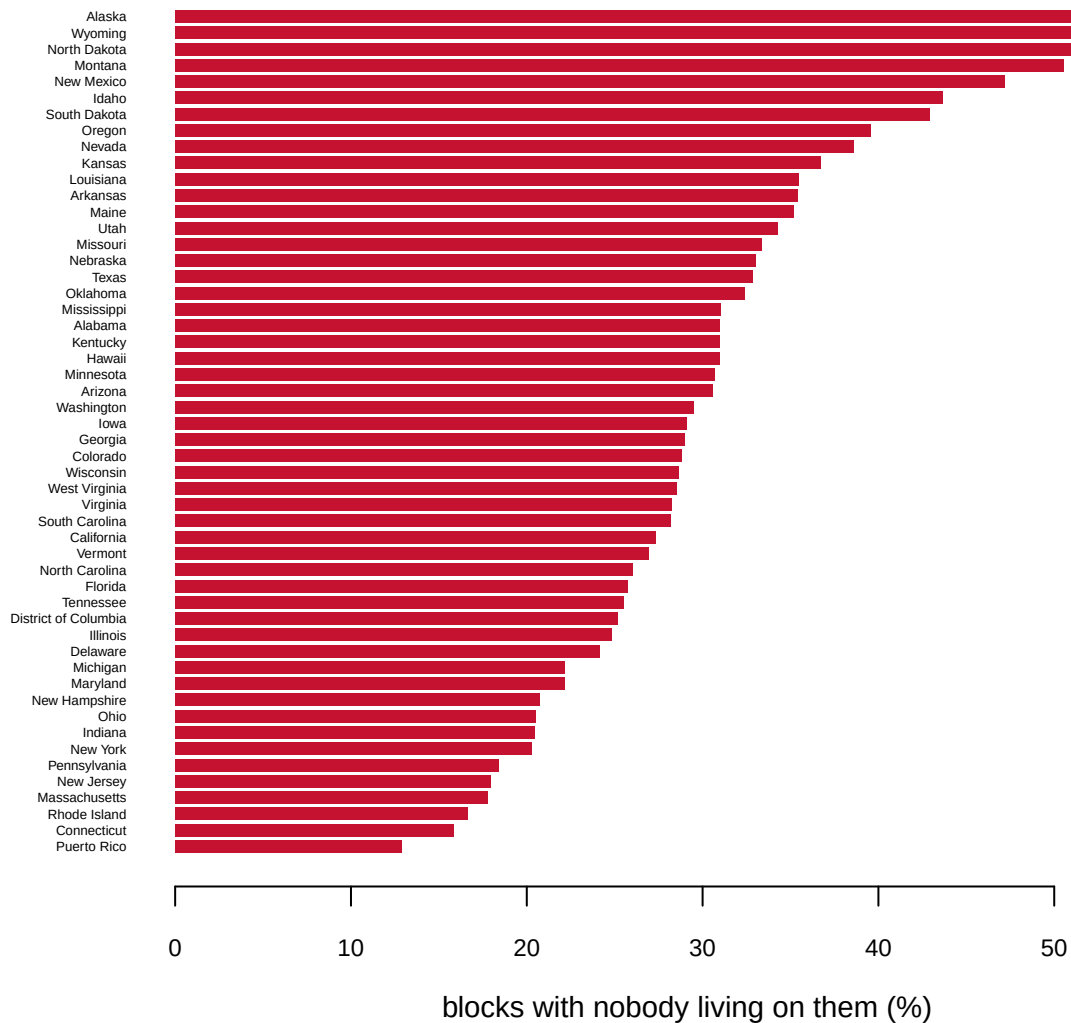


Figure 3. The share of each state's blocks that recorded nobody. Sorted rather than alphabetical, because the ordering is the finding. Hovering a bar in the web edition gives the counts underneath the percentage.

The range runs from Alaska at 58.8% to Puerto Rico at 12.9%. It is tempting to read this as a map of emptiness, and it is partly that. But it is also a map of how the Bureau cut each state. Blocks are drawn along visible features, so a state with a dense grid of section-line roads gets many small blocks whether or not anyone lives on them, and a state with a lot of water gets a lot of water blocks. Alaska is genuinely empty; it also has an enormous coastline and river network, each stretch of which is its own block.

07 Is this just a map of density?

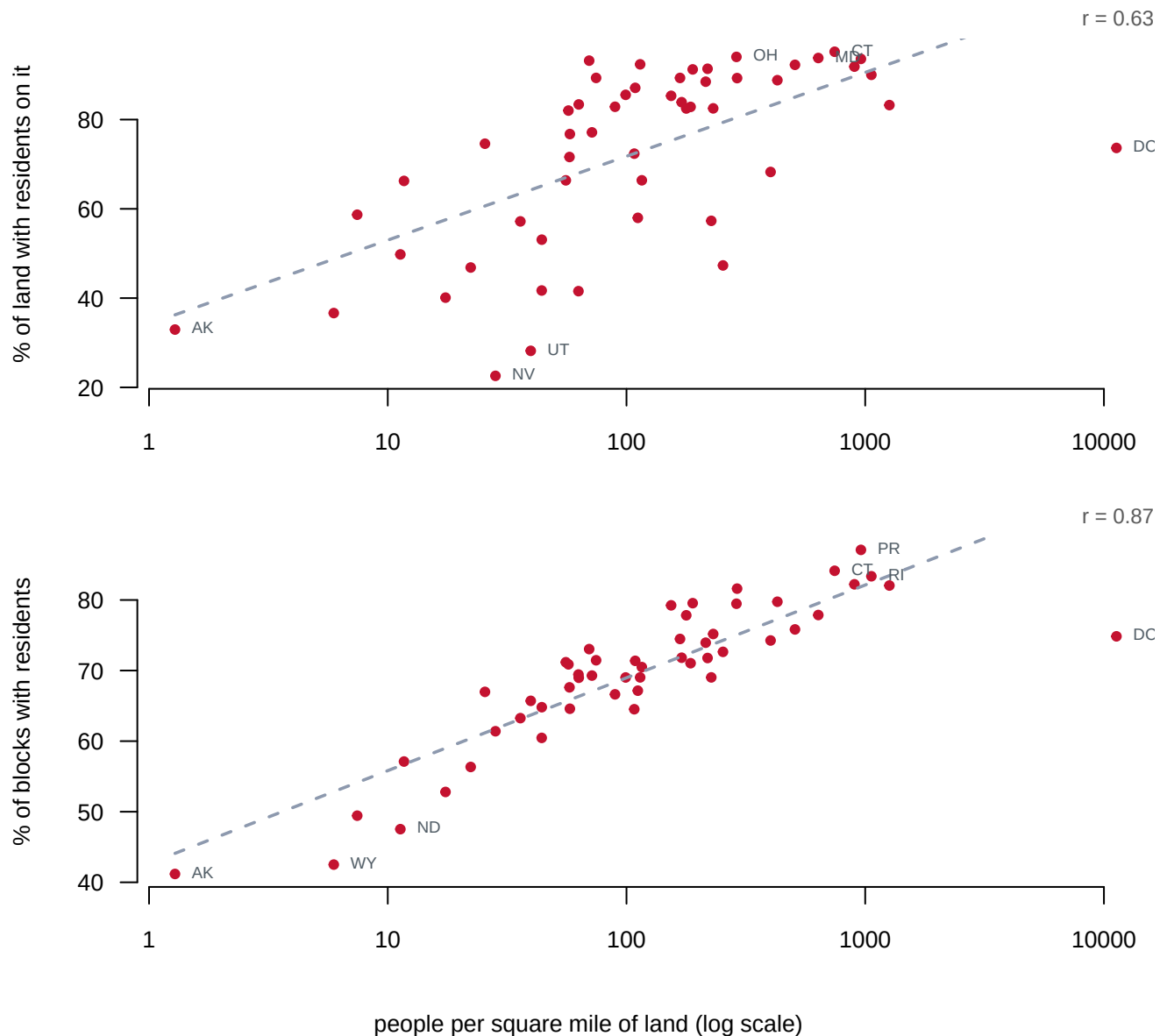


Figure 4. How much of a state is lived on, against how dense it is. One dot per state on a shared log axis, because state densities span four orders of magnitude, from 1.3 people per square mile in Alaska to 11,286 in the District of Columbia. Both panels are drawn as occupancy rather than emptiness, so both rise: the top is the share of a state's *land* with residents on it, the bottom the share of its *blocks*. The chapter's own quantities are one minus each, and the hover gives them. Dashed lines are least-squares fits, stopped where they would leave the panel. Only the extremes and the states furthest from each line are labeled.

The two panels disagree, and the disagreement is the point. Density predicts how much of a state's **blocks** are lived on rather well – it accounts for about 76% of the variation between states. It predicts how much of a state's **land** is lived on much less well: about 39%.

That ordering is backwards from what you would want, and the reason is the thing this brief keeps returning to. A block count is partly a record of how the Bureau cut a state, and the Bureau cuts dense places into small pieces. So the block measure is contaminated by the very thing it is being compared against, and tracks it well for a reason that has little to do with land. The land measure is the substantive one, and it is loose: Nevada has residents on only 22.6% of its land and Connecticut on 95.2%, a difference that is about federal ownership and settlement pattern rather than about people per square mile. The two measures agree with each other only to 0.74.

On the raw scale neither relationship survives; both correlations fall to near nothing. Whatever this tracks, it tracks the *order of magnitude* of a state's density, so the distance from Wyoming to Ohio matters far more than the distance from Ohio to New Jersey.

Part of the answer is the threshold, which is one person. A block counts as inhabited if a single resident is counted anywhere on it, however many acres it covers, and the map says nothing more: white means one resident or a thousand and one, with no way to tell them apart. So a population spread thinly but evenly can leave a state looking full, and a population gathered into one corner can leave the same state looking empty, at the same density either way.

A word first about what "no residents" does and does not mean. It is a statement about where people sleep, not about whether the land is used: a block holding a steel mill, a rail yard, a shopping center or an airport has nobody living on it and is red here. What these maps show is land with no residents, which is a narrower thing than empty land.

Kansas and Utah put it beyond argument. They are the same size — 81,759 and 82,377 square miles, less than one percent apart — and they hold nearly the same number of people, 2,937,880 and 3,271,616, so their densities sit within four people per square mile of each other. Nobody lives on 42.8% of Kansas. Nobody lives on 71.8% of Utah: 24,155 square miles more, which is more ground than the whole of West Virginia.

What separates them is not how many people there are but where the people are allowed to be. Utah's are gathered along the Wasatch Front and most of the rest of the state is federal — range, forest, park, proving ground. Kansas's are spread thinly but evenly across private farmland on a section grid, and a dispersed population is very good at putting at least one person on every block. The block counts carry the same fact: Kansas is cut into 172,529 blocks against Utah's 71,207 for the same ground, averaging 303 acres each against 740.

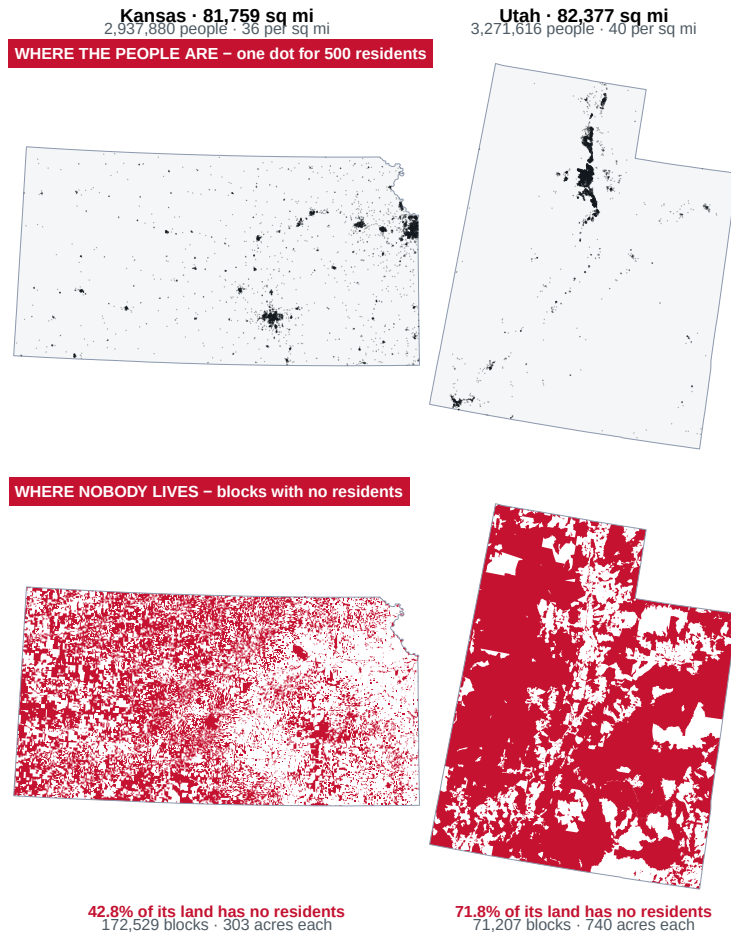


Figure 5. Two states the same size, holding the same number of people, 24,155 square miles apart on how much of them is lived on. Kansas and Utah, four panels at one scale. The top row is population – one dot for 500 residents, scattered inside the block it belongs to rather than stacked on the block's center. The bottom row is the same two states with every block that has no residents in red. Reading down a column shows what a correlation cannot: Utah's people are gathered in one band along the Wasatch Front and the rest of the state is red, while Kansas's are spread over the whole state and its red is a speckle threaded between them.

So what is red on these maps is not thinly settled land. It is land with nobody, and the two are different things.

And then there is the point stranded at the right of both panels.

08 One city, block by block

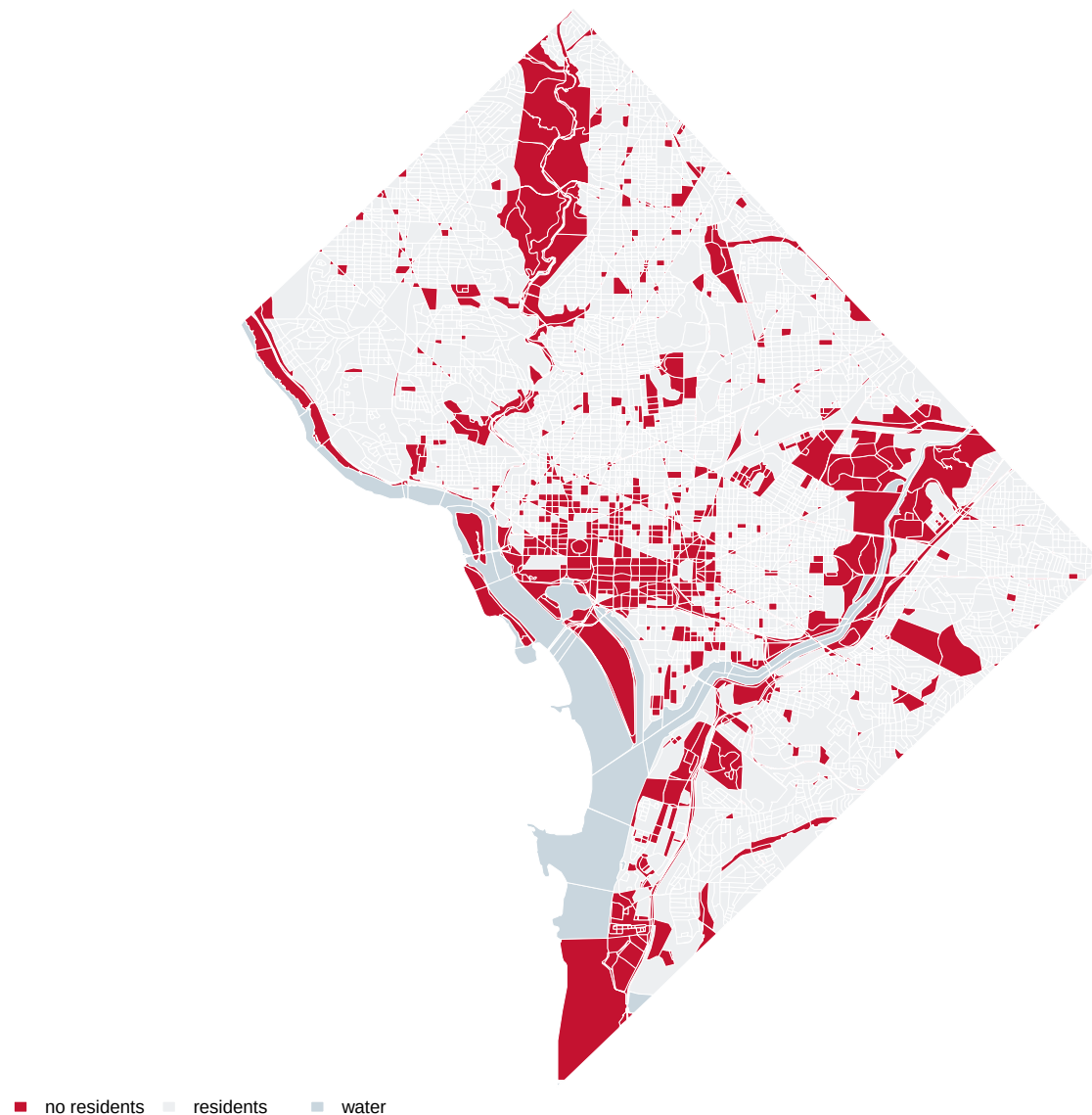


Figure 6. Every census block in the District of Columbia. Red is a block with no residents, pale gray is a block with residents, and blue-gray is a block that is entirely water. Water is drawn as its own class rather than folded into the red, because a block with nobody on it *because it is the Potomac* is a different fact about a city than a block with nobody on it because it is a park, and merging the two would let the rivers argue the case.

The District is the densest place in the country, 11,286 people per square mile of land. 1,512 of its 6,012 blocks — 25.1% — recorded nobody, and 1,409 of those are dry land. They cover 16.1 of the District's 61.1 square miles, 26.4% of it.

The shapes are recognizable if you know the city, and that is the argument. The long wedge running north from the middle is Rock Creek Park. The band across the center, west of the Capitol, is the Mall. The eastern edge is the Anacostia and the parkland along it; the large form in the south is a military installation. None of this is unpopulated in the sense of being remote. It is unpopulated because it is a park, a monument, a rail yard, a base, a river — land in a dense city that no one is counted as living on.

That is why the District sits where it does in Figure 4: it is not a failure of the relationship between density and emptiness so much as a demonstration of what emptiness is actually measuring. **A place can be as dense as anywhere in America and still be a quarter empty**, because emptiness is a fact about how land is used and how it was cut, not only about how many people live nearby.

09 What this brief cannot tell you

The 29.0% is a fact about blocks, not about people or land, and the three do not track each other. Nothing here says how much of the country is *uninhabitable*, only which units recorded no usual residents on 1 April 2020. A block containing a large prison, a dormitory closed for the pandemic, or a seasonal resort counted empty that day may be busy most of the year.

The map is drawn from the land blocks. 238,349 of the empty blocks are water and are not drawn, because including them fills the Great Lakes and every coastal bay with solid color and makes the map a map of hydrography. That is a presentation choice, and it is reversible; the land-area figures are unaffected either way, because the Bureau's ALAND20 already excludes water.

Empty is not the same as thinly settled, and this is not a population density map, though Figure 4 shows how much of one it is. Density is people divided by area, a continuous quantity every block has some value of. Emptiness is a threshold applied to a unit: zero, or not zero. A solid red county on this map means its blocks crossed that threshold, not that nobody lives nearby.

Simplification means no individual block on this map is the right shape. Do not measure anything off it. The built tables carry the counts; the figure carries the pattern.

And zero is a published number, not a raw one. The 2020 counts passed through disclosure avoidance before release, which can move a small count by a little. At the block level that noise is proportionally largest, though a block that is a reservoir was not going to be populated either way.

10 What you should have learned

- The census block is small: the median holds 14 people, and 45.3% hold ten or fewer. Percentages computed on units that size are fragile.
- 29.0% of blocks are empty but they cover about 41.5% of the land. Counting units and measuring area are different operations, and “most of the country is empty” and “most blocks are empty” are different claims.
- A join that matches 100% is a test you passed, not a formality. Identifiers are text.
- Every step that made this map — filtering, dissolving, simplifying — discarded something. A figure is trustworthy when you can say what it discarded and why, not when nothing was discarded.

11 Extensions

- `state_blocks.csv` — divide pop by blocks for each state. Which states put the fewest people in a block, and is that the same list as the emptiest states in Figure 3?
- `state_blocks.csv` joined to `zero_land.csv` — rank states by `pct_zero`, then by `zero_land_sqmi`. Name a state that is high on one list and low on the other, and say what the Bureau did to produce that gap.
- `block_pop_hist.csv` — the last row is a bucket, every block of 1,000 people or more. How many blocks are in it, and what kind of place holds that many people inside one block?
- `block_pop_hist.csv` — compute the share of the national population living in blocks of ten people or fewer. Compare it with the 45.3% of *blocks* that size. Explain the difference to someone in one sentence.
- **Stretch.** Download one state's block file and check its POP20 against the same counts served by the API. Report how many disagree, and decide which you would trust in print.
- **Stretch.** Rebuild Figure 1 for one state without the dissolve step, drawing each empty block separately. Say what the dissolved version hides and what it clarifies.

12 Sources

Data

- U.S. Census Bureau, TIGER/Line Shapefiles 2020, TABBLOCK20 layer — the boundary of every 2020 tabulation block, one file per state, carrying GEOID20, ALAND20, AWATER20 and POP20. This is the only file the build downloads: POP20 is the 2020 decennial count for that block, so the population arrives prepended to the geography. <https://www2.census.gov/geo/tiger/TIGER2020/TABBLOCK20/>
- U.S. Census Bureau, 2020 Census Redistricting Data (P.L. 94-171) Summary File — the decennial release POP20 reports, and the file state legislatures receive. Not downloaded here; cited because it is what the number in the shapefile is. <https://www.census.gov/programs-surveys/decennial-census/about/rdo/summary-files.html>
- U.S. Census Bureau, Cartographic Boundary Files 2020, state layer at 1:500,000 — generalized outlines drawn for mapping rather than measurement, clipped to the shoreline, used here only as the frame of Figure 1. <https://www.census.gov/geographies/mapping-files/time-series/geo/cartographic-boundary.html>

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`block_pop_hist.csv`, `block_size.csv`, `dc_blocks.csv`, `dc_facts.csv`, `dc_frame.csv`, `facts.csv`, `map_frame.csv`, `map_states.csv`, `map_zero.csv`, `state_blocks.csv`, `zero_land.csv`

Method

- Bloch, M. *mapshaper* — the simplification, dissolve and projection steps in Figure 1 are all mapshaper commands; its Visvalingam implementation is what “2% of the vertices” means here. <https://github.com/mbloch/mapshaper>

- Bostock, M. *D3* — the three figures in the HTML edition are plain SVG built with D3 scales and axes. <https://d3js.org/>

Documentation

- U.S. Census Bureau, "Glossary: Census Block" — the definition, and the rule that blocks are bounded by visible features and by legal boundaries. <https://www.census.gov/programs-surveys/geography/about/glossary.html>
- U.S. Census Bureau, "Disclosure Avoidance and the 2020 Census" — why a published block count is not a raw block count. <https://www.census.gov/programs-surveys/decennial-census/decade/2020/planning-management/process/disclosure-avoidance.html>

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original source, without any starting files. Please do the following and show your work.

THE SOURCE, and there is only one. TIGER/Line 2020 tabulation blocks, one

zipped shapefile per state:

https://www2.census.gov/geo/tiger/TIGER2020/TABBLOCK20/tl_2020_<FIPS>_tabblock20.zip

Open, no key, no account. About 10 GB for the whole country across 52 files

(the 50 states, DC and Puerto Rico).

Every block record carries its own population. POP20 is the total population

the 2020 decennial census counted in that block — a complete count, not an

estimate and not a sample, the same number published in the P.L. 94-171 redistricting file. It comes prepended to the geography, so the boundary and

the count arrive in one file and nothing has to be joined. Do not fetch the

Census API for this: it serves the same counts one variable at a time and

will not return a whole state's blocks at once, so it turns a 52-file download into some 3,100 requests.

The attribute table also carries GEOID20 (the 15-character block identifier),

ALAND20 (land area, square meters) and AWATER20. For the tables below you

need only the attribute table, the .dbf inside each zip; the geometry is

needed only for the map.

WHAT TO BUILD.

1. A table of every census block in those 52 files with its 2020 population.

Report the national total and how many blocks recorded zero.

2. A count of blocks at each exact population from 0 to 1,000, with everything

above 1,000 in one bucket. Report the median block population, the median

among blocks with at least one resident, and the share holding ten or fewer.

3. A per-state table: blocks, blocks with nobody, total land area, and the

percentage of each.

4. A national map of the zero-population blocks. Keep the blocks where POP20

is zero, and COUNT THEM BEFORE YOU CHANGE THE GEOMETRY: how many there are, how many of them have land area (the rest are water), and how many square miles of land they cover. Only then dissolve the touching ones together for drawing.

THINGS THAT WILL BITE YOU.

– GEOID20 is text, not a number. Read as a number, a 15-digit identifier becomes 1.3121e+14 and leading zeros vanish. The same is true of the 2-character state code: "09" read as 9 will not match "09".

– Water blocks have zero population and are real. Decide whether they belong

on your map and say which you chose; drawing them fills the Great Lakes.

– Dissolving destroys the block as a unit. After it, counting shapes counts

REGIONS, not blocks: a thousand touching empty blocks become one polygon.

Take every number about blocks off the undissolved table, or carry it through the dissolve as a summed field – give each block a 1 and sum it, and sum ALAND20 the same way. If your block count drops when you dissolve,

you are reporting the wrong thing.

– Dissolving makes polygons with holes. Draw each polygon's rings together

under an even-odd fill rule, or the holes get painted as fill.

- Round coordinates only AFTER projecting. Rounding longitude and latitude

rounds to whole degrees.

- Simplify each state to the size it will be DRAWN at, not to one national

tolerance, if you want a map that survives being zoomed into.

CHECK YOUR WORK. The copies this book used were downloaded in September 2026. If your rebuild matches them, you should find:

- 8,174,955 blocks holding 334,735,155 people, across 52 states and state-equivalents

- 2,368,443 blocks with nobody living on them – 29.0%

- a median block population of 14, or 28 among blocks with at least one resident; 45.3% of blocks hold ten people or fewer

- emptiest Alaska at 58.82%, least empty Puerto Rico at 12.90%

- 2,130,094 of the empty blocks have land area, covering 1,465,387 square miles – about 41% of U.S. land, against 29% of blocks

- the largest block by area is 9,226 square miles in the North Slope Borough, Alaska, and its population is zero

TIGER/Line 2020 is a finished product; it will not change until the 2030 cycle publishes its successor.